HiveWatch: framework-agnostic FL observability

A server-side observability layer that monitors FL execution without changing framework-controlled training, communication, or aggregation

① FEDERATED WORKLOAD

Federated clients

Private data + local training

Client A
model +
metadata (metrics)

Client B
model +
metadata (metrics)

Client N
model +
metadata (metrics)

model +
metadata

any transport

FL server

framework-owned

- Client selection
- Transport
- Scheduling
- Aggregation
- Evaluation

`APPFL · Flower`
`NVIDIA FLARE ·`
Others

API calls

② HIVEWATCH CORE

HiveWatch Runtime

Minimal Runtime APIs

```
init() · round_start() ·
log_client_update()
log_round() · finish()
```

FL-native Event Schema

rounds · clients · failures · checkpoints
communication · resources · geography

Normalization + Derived Metrics

duration · participation · byte totals
staleness · gradient-norm dispersion

pluggable emitter dispatch

③ PLUGGABLE EMITTERS

GeoEmitter

Append-only JSONL event history

Map-ready snapshot

live SSE stream | dashboard + replay

WandBEmitter

Per-client and round metrics

Hosted dashboards · artifacts · tables

MLflowEmitter

Local or remote tracking server

Metrics · parameters · tags · checkpoints

④ ANALYSIS





